Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2000-II-Q05] | Part A (Q1-Q5)

Answer: 376

1. Choose five rings in $\binom{8}{5}$ ways and order them in $5!$ ways, reading fingers in a fixed order.

2. The four finger lengths form a weak composition of 5, with $\binom{8}{3}$ choices. Thus $N = 56 \cdot 120 \cdot 56 = 376320$, whose requested digits are 376.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2021-I-Q04] | Part A (Q1-Q5)

Answer: 331

1. There are $\binom{65}{2} = 2080$ ordered positive triples. One has all entries equal.

2. For exactly two equal entries, use the multiset $\{u, u, 66 - 2u\}$ with $1 \leq u \leq 32$ and $u \neq 22$. The 31 multisets have three orderings each, giving 93.

3. The remaining $2080 - 1 - 93 = 1986$ triples have distinct entries. Each increasing triple has six orderings, so the answer is $1986/6 = 331$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2003-I-Q09] | Part B (Q6-Q10)

Answer: 615

1. For common digit sum $1 \leq s \leq 9$, the first pair has s possibilities because its first digit is nonzero; the last pair has $s+1$. For $10 \leq s \leq 18$, both pairs have $19-s$ possibilities.

2. The count is $\sum_{s=1}^{9} s(s+1) + \sum_{s=10}^{18} (19-s)^2 = 2\sum_{k=1}^{9} k^2 + \sum_{k=1}^{9} k = 615$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q09] | Part B (Q6-Q10)

Answer: 420

1. Arrange the seven digits in $7!/(4!3!) = 35$ ways. Then insert two cuts to form three nonempty blocks, each length at most four.

2. Allowed length patterns are permutations of $(1, 2, 4), (1, 3, 3), (2, 2, 3)$, with counts $6, 3, 3$. No leading-zero issue occurs. Thus $35 \cdot 12 = 420$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q10] | Part B (Q6-Q10)

Answer: 346

1. Ignoring individuals, group patterns consist of k ordered triples of nonempty M, V, E runs, for $1 \leq k \leq 5$. The seat-one/seat-fifteen condition fixes both endpoints at run boundaries.

2. For each group, split its five occurrences into k ordered positive run lengths, giving $\binom{4}{k-1}$. The groups are independent, so $N = \sum_{j=0}^{4} \binom{4}{j}^3 = 1 + 64 + 216 + 64 + 1 = 346$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2011-II-Q07] | Part B (Q6-Q10)

Answer: 003

1. The five greens make at most five green runs and therefore at most six red runs. Thus there are at most ten color changes. With m reds there are $m+4$ adjacent pairs, so equality forces $m+4 \leq 20$ and $m \leq 16$.

2. At $m=16$, all eleven runs must be present: six nonempty red runs separated by the five singleton greens. Any positive composition of sixteen into six parts attains the condition.

3. Stars and bars gives $N=\binom{15}{5}=3003$, with remainder 3.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2025-II-Q10] | Part B (Q6-Q10)

Answer: 907

1. Occupied runs have length one or two. If there are j runs, then $8-j$ are doubletons and $2j-8$ are singletons, so $4 \le j \le 8$. Their order has $\binom{j}{8-j}$ choices.

2. The eight empty chairs create nine gaps, of which choose j for these runs. Thus $N = \sum_{j=4}^{8} \binom{j}{8-j}\binom{9}{j} = 126 + 1260 + 1260 + 252 + 9 = 2907$. The remainder is 907.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2008-II-Q12] | Part C (Q11-Q15)

Answer: 310

1. If pole one has b blues and g greens, its arrangements number $\binom{b+1}{g}$ using distinct gaps around the blues. Pole two contributes $\binom{11-b}{9-g}$.

2. Allow empty poles temporarily and sum: Vandermonde gives $\sum_{b=0}^{10}\sum_{g}\binom{b+1}{g}\binom{11-b}{9-g} = 11\binom{12}{9} = 2420$.

3. An empty pole leaves all flags on the other in $\binom{11}{9} = 55$ ways, for each of two choices. Thus $N = 2420 - 110 = 2310$, remainder 310.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2018-II-Q15] | Part C (Q11-Q15)

Answer: 185

1. Let $d_i = f(i) - f(i-1) \in \{\pm 1, \pm 2, \pm 3\}$. Their sum is twelve. Two negative steps would make the sum at most $18 - 8 = 10$, so at most one is negative.

2. With no negatives, all pair inequalities follow immediately from steps one through three. The counts of ones, twos, threes can be $(0, 6, 0), (1, 4, 1), (2, 2, 2), (3, 0, 3)$, contributing $1 + 30 + 90 + 20 = 141$.

3. A negative two cannot adjoin any allowed positive step without violating the distance-two lower bound. A negative three would force all five other steps to be three to achieve total twelve, again violating that bound. Thus a single negative must be minus one, and its existing neighbors must be three. These local conditions suffice for every longer interval: a two-step interval containing it has sum at least two, and extending by positive steps preserves the lower bound; intervals containing it internally are larger still.

4. At either endpoint, the remaining four unfixed positive steps sum to ten, giving $4 + 6 = 10$ orders each. At each of four interior positions, the remaining three positive steps sum to seven, giving $3 + 3 = 6$ orders each. Total $141 + 20 + 24 = 185$.

Source / 来源

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2022-II-Q13] | Part C (Q11-Q15)

Answer: 220

1. The numerator has constant term 1 and no other terms below degree 2310. Expanding the four geometric series, the desired coefficient counts $105u + 70v + 42w + 30t = 2022$ with all variables nonnegative integers.

2. Reduce modulo $2, 3, 5, 7$: $u = 2u_0, v = 3v_0, w = 5w_0 + 1, t = 7t_0 + 3$. These substitutions are forced, and conversely preserve nonnegativity.

3. The equation becomes $210(u_0 + v_0 + w_0 + t_0) + 132 = 2022$, or sum 9. Stars and bars gives $\binom{12}{3} = 220$.

Source / 来源